

EC200U&EG91xU Series

QuecOpen(SDK) vSIM Development Guide

LTE Standard Module Series

Version: 1.0

Date: 2023-10-30

Status: Released



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Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

Or our local offices. For more information, please visit:

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About the Document

Revision History

Version	Date	Author	Description
-	2023-08-03	Joe TU	Creation of the document
1.0	2023-10-30	Joe TU	First official release

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1 Introduction

Quectel EC200U series and EG91xU family modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS. It is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see **document [1]**.

This document is applicable to QuecOpen® solution based on CSDK build environment, and introduces vSIM adaptation layer API of EC200U series and EG91xU family modules, including API calling process, API details and API examples.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200U Series
EG91xU	EG912U-GL
	EG915U Series

2 API Calling Process

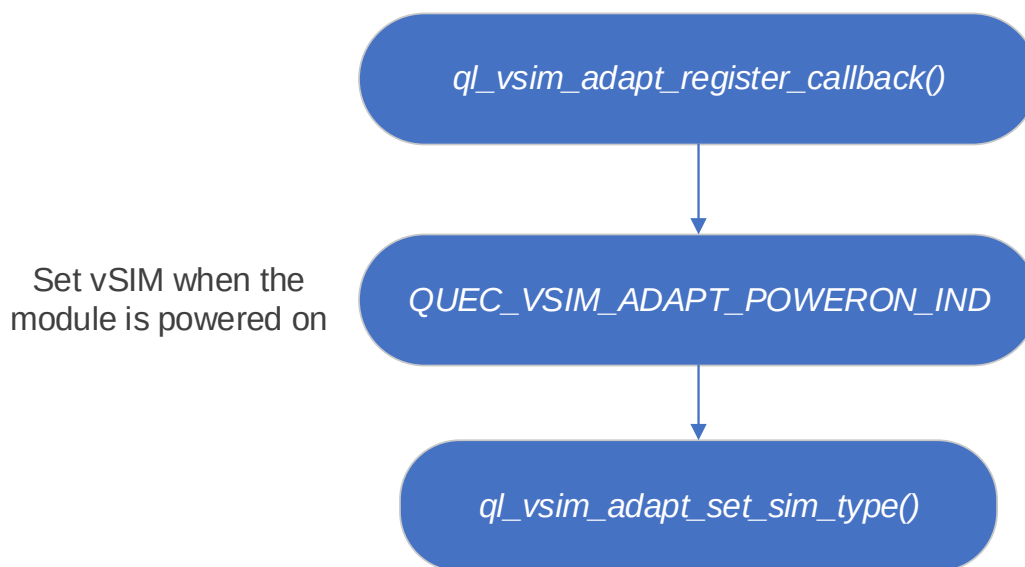


Figure 1: API Calling Process

The figure above shows the process of setting the type of SIM card when the module is powered on, and the relevant sample code is in the *components\appstart\src\app_start_XXX.c* directory.

3 vSIM API

3.1. Header File

ql_api_vsim_adapt.h, the header file of vSIM adaptation layer API, is in the *components\ql-kernel\inc* directory of the CSDK. Unless otherwise specified, all the header files mentioned in this document are in this directory.

3.2. API Overview

Table 2: API Overview

Function	Description
<i>ql_vsim_adapt_register_callback()</i>	Registers the vSIM adaptation layer callback function.
<i>ql_vsim_adapt_set_sim_type()</i>	Sets the vSIM adaptation layer SIM card type and the callback functions of APDU and ATR interface related to vSIM.
<i>ql_vsim_adapt_get_sim_type()</i>	Gets the SIM card type of the vSIM adaptation layer.

3.3. API Description

3.3.1. ql_vsim_adapt_register_callback

This function registers the vSIM adaptation layer callback function. Please register the callback function as soon as possible after booting, otherwise the *QUEC_VSIM_ADAPT_POWERON_IND* event may not be obtained.

● Prototype

```
ql_vsim_adapt_errcode_e ql_vsim_adapt_register_callback(ql_vsim_adapt_callback vsim_adapt_cb)
```

- **Parameter**

vsim_adapt_cb:

[In] Callback function of vSIM adaptation layer related events. See **Chapter 3.3.1.1** for details.

- **Return Value**

See **Chapter 3.3.1.2** for details.

3.3.1.1. ql_vsim_adapt_callback

This callback function receives events related to the vSIM adaptation layer.

- **Prototype**

```
typedef int (*ql_vsim_adapt_callback)(uint32_t ind_type, void *ctx)
```

- **Parameter**

ind_type:

[In] vSIM adaptation layer event ID.

QUEEC_VSIM_ADAPT_POWERON_IND event notifies the switching of SIM card type during power-on. If you need to switch the SIM card type to vSIM during power-on, you need to call the *ql_vsim_adapt_set_sim_type()* to switch the SIM card type to *QL_VSIM_ADAPT_SIM_TYPE_SSIM* or *QL_VSIM_ADAPT_SIM_TYPE_CSIM* after receiving this event. If this event is ignored, the regular physical SIM card will be initialized at booting.

ctx:

[In] Event context.

- **Return Value**

None

3.3.1.2. ql_vsim_adapt_errcode_e

The error code of the vSIM adaptation layer API, indicating whether the function is executed successfully. If the execution fails, the reason is returned, and the enumeration is defined below:

```
typedef enum
{
    QL_VSIM_ADAPT_SUCCESS                = 0,
    QL_VSIM_ADAPT_ERROR                  = 1 | (QL_COMPONENT_VSIM_ADAPT << 16),
    QL_VSIM_ADAPT_PARA_ERR,
```

```
}ql_vsim_adapt_errcode_e
```

● Member

Member	Description
<code>QL_VSIM_ADAPT_SUCCESS</code>	Successful execution.
<code>QL_VSIM_ADAPT_ERROR</code>	Failed execution.
<code>QL_VSIM_ADAPT_PARA_ERR</code>	Parameter error.

3.3.2. ql_vsim_adapt_set_sim_type

This function sets the vSIM adaptation layer SIM card type and the callback functions of APDU and ATR interface related to vSIM.

● Prototype

```
ql_vsim_adapt_errcode_e ql_vsim_adapt_set_sim_type(ql_vsim_adapt_sim_type_e sim_type,const
ql_vsim_adapt_handler_s *handler, uint8_t nSim)
```

● Parameter

sim_type:

[In] SIM card type. See **Chapter 3.3.2.1** for details.

handler:

[In] The APDU and ATR interface implemented by vSIM developers. When the SIM card is set to a non-physical card, the APDU and ATR features need to be implemented by the vSIM developer. See **Chapter 3.3.2.2** for details.

nSim:

[In] The (U)SIM card used. Set this parameter to 0 if the module supports only one (U)SIM interface.

0 (U)SIM card 1

1 (U)SIM card 2

● Return Value

See **Chapter 3.3.1.2** for details.

3.3.2.1. ql_vsim_adapt_sim_type_e

The enumeration of SIM card type of vSIM adaptation layer is defined below:

```
typedef enum
{
    QL_VSIM_ADAPT_SIM_TYPE_TSIM,        //Physical SIM card
    QL_VSIM_ADAPT_SIM_TYPE_SSIM,        //SoftSIM card
    QL_VSIM_ADAPT_SIM_TYPE_CSIM,        //Cloud SIM card
    QL_VSIM_ADAPT_SIM_TYPE_MAX,
} ql_vsim_adapt_sim_type_e
```

● Member

Member	Description
<i>QL_VSIM_ADAPT_SIM_TYPE_TSIM</i>	Physical SIM card.
<i>QL_VSIM_ADAPT_SIM_TYPE_SSIM</i>	SoftSIM card.
<i>QL_VSIM_ADAPT_SIM_TYPE_CSIM</i>	Cloud vSIM card. It needs to be used in conjunction with Dual SIM Dual Standby technology. During network registration, the cloud vSIM card relies on the seed card to perform authentication and calculation in the Cloud.

3.3.2.2. ql_vsim_adapt_handler_s

The APDU and ATR interface implemented by vSIM developers. When the SIM card type is set to *QL_VSIM_ADAPT_SIM_TYPE_SSIM* or *QL_VSIM_ADAPT_SIM_TYPE_CSIM* in the kernel, the APDU and ATR interface is used for APDU and ATR communication with the kernel.

The structure of APDU and ATR interface is defined below:

```
typedef struct
{
    ql_vsim_adapt_process_apdu process_apdu;
    ql_vsim_adapt_reset reset;
}ql_vsim_adapt_handler_s
```

● Parameter

Type	Parameter	Description
<i>ql_vsim_adapt_process_apdu</i>	<i>process_apdu</i>	APDU interface implemented by vSIM developers. See Chapter 3.3.2.3 for details.
<i>ql_vsim_adapt_reset</i>	<i>reset</i>	ATR interface implemented by vSIM developers. See Chapter 3.3.2.4 for details.

3.3.2.3. ql_vsim_adapt_process_apdu

This function is for implementing vSIM APDU.

- **Prototype**

```
typedef int (*ql_vsim_adapt_process_apdu)(uint8_t *apdu_req, uint16_t apdu_req_len, uint8_t *apdu_rsp, uint16_t *apdu_rsp_len, uint8_t slot)
```

- **Parameter**

apdu_req:

[In] Command APDU.

apdu_req_len:

[In] The length of command APDU.

apdu_rsp:

[Out] Response APDU.

apdu_rsp_len:

[Out] The length of response APDU.

slot:

[In] The (U)SIM card used. Set this parameter to 0 if the module supports only one (U)SIM interface.

0 (U)SIM card 1

1 (U)SIM card 2

- **Return Value**

SW1-SW2 status word.

3.3.2.4. ql_vsim_adapt_reset

This function is the vSIM ATR implementation function.

- **Prototype**

```
typedef uint16_t (*ql_vsim_adapt_reset)(uint8_t *atr_data, uint8_t *atr_size, uint8_t nSim)
```

- **Parameter**

atr_data:

[Out] The returned ATR data.

atr_size:

[Out] The length of the returned ATR data.

nSim:

[In] The (U)SIM card used. Set this parameter to 0 if the module supports only one (U)SIM interface.

0 (U)SIM card 1

1 (U)SIM card 2

- **Return Value**

0 Successful execution

Other values Failed execution

3.3.3. ql_vsim_adapt_get_sim_type

This function gets the SIM card type of the vSIM adaptation layer.

- **Prototype**

```
ql_vsim_adapt_sim_type_e ql_vsim_adapt_get_sim_type(uint8_t nSim)
```

- **Parameter**

nSim:

[In] The (U)SIM card used. Set this parameter to 0 if the module supports only one (U)SIM interface.

0 (U)SIM card 1

1 (U)SIM card 2

- **Return Value**

See **Chapter 3.3.2.1** for details.

4 Example

The example of vSIM adaptation layer interfaces is in the *components\ql-application\vsim_adapt\vsim_adapt_demo.c* directory of the CSDK; the sample code for vSIM adaptation layer event reception processing and SIM type setting at booting are in the *components\appstart\src\app_start_xxx.c* directory.

5 Appendix References

Table 3: Related Document

Document Name
[1] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Quick_Start_Guide

Table 4: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
APDU	Application Protocol Data Unit
ATR	Answer To Reset
DSDS	Dual SIM Dual Standby
SDK	Software Development Kit
SIM	Subscriber Identity Module
SW1-SW2	Status Word1 – Status Word2
(U)SIM	(Universal) Subscriber Identity Module
vSIM	virtual-SIM